EBIKE101 Day 1 Master Print Packet

Introduction, Rider Safety, and System Inspection

CTS Cohort 6 | Day 33 | Monday, August 31, 2026

Safe micro-mobility starts before the throttle moves.

Instructor Run Sheet

EBIKE101 | CTS Cohort 6 | Day 33

EBIKE101 Day 1: Introduction, Rider Safety, and System Inspection

Safe micro-mobility starts before the throttle moves.

Official focus

Students begin the Electric Micro Mobility block by treating an e-bike as a vehicle system, not a toy. Day 1 introduces rider safety, motor/throttle control boundaries, the difference between conventional bicycle systems and electric assist systems, and the pre-ride inspection standard students will use for the full two-week section.

The nearly completed EDM101 subwoofer may be shown as a class culture carryover only. It is not part of the official EBIKE101 curriculum, grading, or learning targets.

Today's big question

What makes an e-bike different from a bicycle, and what must be proven safe before anybody rides it?

Core rule

No ride, no throttle test, and no powered wheel spin until the bike passes a visible safety check.

Learning targets

Students will be able to:

1. Identify the main systems on a conventional bicycle: frame, fork, wheels, tires, brakes, drivetrain, cockpit, and contact points.

2. Identify the main e-bike systems: battery, BMS concept, motor, controller, display, throttle or pedal assist sensor, wiring, connectors, charger, and cutoffs.

3. Explain why rider control, braking, tire condition, wiring security, and battery condition must be checked before operation.

4. Complete a pre-ride inspection checklist and record pass/fail notes.

5. Trace the basic e-bike flow: rider input -> controller decision -> battery power -> motor output -> wheel motion.

Readiness and materials

Before students arrive

- Stage one e-bike or conversion bike where students can walk around it.
- Stage one conventional bike if available for comparison.
- Keep battery disconnected or system powered off until the safety boundary is taught.
- Mark a no-ride/no-throttle area around the bike.
- Print student packets and quiz sheets.
- Put inspection categories on the board:

- mechanical

- electrical

- rider control

- route/environment

- documentation

Materials

- E-bike or conversion bike
- Conventional bike, if available
- Bike stand, if available
- Helmets
- Gloves and eye protection as appropriate
- Tire pump and pressure gauge
- Basic bike tools: hex keys, wrench set, tire levers
- Multimeter for demonstration only
- Cones or tape for controlled area planning
- Zip ties or Velcro straps for loose-wire examples
- Printed student packets, quiz, answer key, and lab sheets

Five-hour run of show

Time	Block	Instructor move
10:00-10:10	Morning landing	Attendance, settle, phones down, quick context: EBIKE101 starts today. Ask: "What can go wrong if a powered bike is treated like a normal bike?"
10:10-10:30	Morning quiz	20-question readiness quiz. Grade for safety thinking and vocabulary, not punishment. Fix unsafe assumptions immediately.
10:30-10:50	Course launch	Explain the two-week arc: inspect -> understand -> plan -> convert -> wire -> troubleshoot -> commission -> certify.
10:50-11:15	Bicycle baseline	Walk around a normal bike. Identify frame, fork, wheels, tires, brakes, chain/drivetrain, handlebars, seat, and contact points.
11:15-11:45	E-bike added systems	Add battery, controller, motor, display, throttle/PAS, wiring, charger, cutoffs, and connectors. Keep the bike powered off.
11:45-12:00	Safety boundary	Teach no ride/no throttle/no spin until the pre-ride checklist passes. Discuss stop-work authority.
12:00-12:30	Lunch/reset	Keep bikes staged. Reset tools and packets.
12:30-1:10	Guided inspection demo	Instructor models the pre-ride inspection: tires, brakes, steering, axle/wheel security, drivetrain,

		battery mount, wiring, connectors, controls.
1:10-1:55	Lab A: team inspection	Teams inspect bike systems using the student checklist. They mark pass/fail/needs instructor review.
1:55-2:10	Break	Reset, hydrate, keep students out of the no-ride area.
2:10-2:40	Lab B: system map	Teams draw the e-bike flow from rider input to motor output and name what could fail at each step.
2:40-2:55	Debrief	Each team reports one mechanical risk, one electrical risk, and one rider-control risk.
2:55-3:00	Closeout	Collect packets. Preview Day 34: conventional bicycle mechanics, because electricity does not fix loose wheels.

Instructor notes

- Do not start with speed. Start with control.
- Do not let "it turns on" become "it is safe." Those are different claims.
- If the bike is incomplete or unsafe, the lab still works as an inspection and planning exercise.
- If students are excited about riding, use that energy but hold the line: inspection first.
- A bike can be mechanically unsafe even if the electrical system is perfect.
- A bike can be electrically unsafe even if it pedals like a normal bike.
- Do not use public-road law details as the lesson unless verified separately. This class is about controlled training use and safe inspection habits.

Stop-work triggers

Pause the lab if any of these happen:

- A student attempts to ride, throttle, or spin the powered wheel without instruction.
- The bike tips, falls, or becomes unstable in the stand.
- Battery damage, swelling, heat, odor, water intrusion, or unknown charger condition is observed.
- Brakes fail a squeeze/roll check.
- Wheel/axle security is uncertain.
- Wiring can contact moving parts, tire, chain, brake rotor, or crank.
- A student jokes about defeating safeties, bypassing cutoffs, or "just sending it."
- A team cannot explain the next action safely.

Completion standard

By the end of Day 1, each student should be able to say:

"Before operating an e-bike, I must verify the mechanical bike, the electrical assist system, the rider controls, and the riding area. Power only comes after control is proven."

Backup plan

If no e-bike is available today:

1. Use a conventional bike as the physical inspection object.

2. Draw or project an e-bike block diagram.

3. Have teams label where the battery, motor, controller, display, throttle/PAS, charger, and cutoffs would fit.

4. Complete the same pre-ride checklist with "not installed" marked where appropriate.

5. Finish with a paper-based hazard analysis.

Morning Quiz - Student

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EBIKE101 Day 1 Morning Quiz

Rider safety, e-bike systems, and pre-ride thinking

Date: Monday, August 31, 2026

Time: 15-20 minutes

Directions: Choose the best answer. If unsure, choose the safest professional action.

Name: _____ Team: _____

1. What is the first mindset for working around an e-bike?

A. It is just a toy

B. It is a vehicle system with stored energy and moving parts

C. It is safe if the display turns on

D. It is safe if the battery is expensive

2. Which item belongs to a conventional bicycle system?

A. Frame

B. Controller

C. Display

D. Charger

3. Which item belongs to an e-bike electrical system?

A. Battery

B. Pedal

C. Seat

D. Handlebar grip

4. What should happen before any throttle test or powered wheel spin?

A. A video should be recorded

B. The bike should pass a visible safety inspection

C. The fastest student should try it

D. The battery should be fully charged only

5. Why are brakes especially important on an e-bike?

A. E-bikes do not use brakes

B. The bike may be heavier and faster than a normal bike

C. Brakes only matter downhill

D. Brakes are only for legal reasons

6. A loose battery mount is:

A. Fine if the bike still turns on

B. A safety concern because the battery can move, fall, or damage wiring

C. Only a cosmetic problem

D. A problem only during rain

7. Wiring should be routed so it:

A. Can touch the tire

B. Can wrap around the crank

C. Is protected from moving parts, sharp edges, heat, and pinch points

D. Hangs loose for easy troubleshooting

8. What does a controller do in a basic e-bike system?

A. Stores air pressure

B. Converts sunlight to motion

C. Manages power sent from the battery to the motor based on rider input

D. Inflates tires

9. What is rider input?

A. Any command or action from the rider, such as throttle or pedal assist

B. The color of the frame

C. The brand of helmet

D. The tire tread pattern

10. What is a pre-ride inspection?

A. A quick safety check before using the bike

B. A decoration step

C. A battery charging trick

D. A way to skip repairs

11. If tire pressure is very low, the safest action is to:

A. Ride slowly and hope

B. Inflate and inspect before riding

C. Add more battery power

D. Ignore it if the motor works

12. If the wheel does not spin freely because something rubs, the safest action is to:

A. Add throttle to force it

B. Stop and inspect the wheel, brake, tire, and routing

C. Remove the brake

D. Ride only in a straight line

13. If a charger is unknown, damaged, or mismatched, what should happen?

A. Use it quickly

B. Do not use it until it is verified by the instructor

C. Plug it in outside

D. Tape over the damage

14. What does "stop-work authority" mean?

A. Only the instructor can notice hazards

B. Anyone can call a pause when something seems unsafe

C. Work stops only after something breaks

D. It means the class is over

15. Which is the best technical statement?

A. "It looks fine."

B. "I checked tire pressure, brake function, battery mount, wiring clearance, and control response."

C. "It probably works."

D. "Someone else rode it last week."

16. Why should a bike be stable before inspection?

A. So it looks professional only

B. So it does not fall, pinch, damage parts, or injure someone

C. So the battery charges faster

D. So the motor gets stronger

17. What is the safest response to an unknown electrical condition?

A. Assume it is safe

B. Bypass the unknown part

C. Pause, identify the system, and verify before touching or operating

D. Ask the newest student to try it

18. What is the basic e-bike energy path?

A. Helmet -> tire -> seat -> battery

B. Rider input -> controller -> battery power -> motor -> wheel motion

C. Tire pressure -> display -> helmet -> chain

D. Brake lever -> seat -> charger -> pedal

19. What is one reason documentation matters?

A. It makes the bike heavier

B. It helps the next person understand what was inspected, changed, or found unsafe

C. It replaces inspection

D. It makes batteries last forever

20. Today's main lesson is:

A. If it turns on, ride it

B. Power only comes after control is proven

C. E-bikes do not need normal bike maintenance

D. Speed is the first goal

Morning Quiz - Answer Key

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EBIKE101 Day 1 Morning Quiz Answer Key

Use this key to check for safe thinking. Correct unsafe assumptions immediately before lab work.

1. B - An e-bike is a vehicle system with stored energy and moving parts.

2. A - Frame is part of the conventional bike system.

3. A - Battery is part of the e-bike electrical system.

4. B - A visible safety inspection comes before throttle or powered motion.

5. B - E-bikes may be heavier and faster, so braking matters more.

6. B - A loose battery can move, fall, or damage wiring.

7. C - Wiring must be protected from moving parts, sharp edges, heat, and pinch points.

8. C - The controller manages battery power to the motor based on rider input.

9. A - Rider input includes throttle, pedal assist, braking, and control actions.

10. A - A pre-ride inspection is a safety check before use.

11. B - Inflate and inspect before riding.

12. B - Stop and inspect the wheel, brake, tire, and routing.

13. B - Do not use an unknown, damaged, or mismatched charger.

14. B - Anyone can call a safety pause.

15. B - A professional statement names what was checked.

16. B - Stability prevents falls, pinches, damage, and injury.

17. C - Pause, identify, and verify.

18. B - Rider input -> controller -> battery power -> motor -> wheel motion.

19. B - Documentation helps the next person understand the system and the inspection.

20. B - Power only comes after control is proven.

Quick reteach prompts

If students miss questions 4, 12, 13, or 17, reteach before any hands-on work:

- "Turning on is not the same as safe."
- "Motion is not a test until the system is controlled."

- "Unknown electrical condition means pause, identify, verify."
- "No charger, battery, or wiring shortcuts."

Student Print Packet

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EBIKE101 Day 1 Student Packet

Safe micro-mobility starts before the throttle moves.

Name: _____ Team: _____

Today's big question

What makes an e-bike different from a bicycle, and what must be proven safe before anybody rides it?

Part 1: Bicycle baseline

Label or describe each part. Use plain language.

Part	What does it do?	What could make it unsafe?
Frame		
Fork		
Front wheel		
Rear wheel		
Tires		
Brakes		
Chain/drivetrain		
Handlebars/stem		
Seat/seatpost		
Pedals/cranks		

Part 2: E-bike added systems

E-bike part	Job	What could fail?
Battery		
Battery mount		
Motor		
Controller		
Display		
Throttle		
Pedal assist sensor		
Wiring harness		
Connectors		
Charger		
Brake cutoffs, if installed		

Part 3: Basic system flow

Fill in the flow.

Rider input -> _____ -> battery power -> _____ -> wheel motion

What does the rider control?

_____

What does the controller control?

_____

What does the motor move?

_____

Part 4: Pre-ride inspection checklist

Mark each item.

P = Pass

F = Fail

R = Needs instructor review

N/A = Not installed or not available today

Mechanical bike

Item	P/F/R/N/A	Notes
Frame has no obvious cracks, damage, or loose parts		
Fork looks straight and secure		
Wheels are secure		
Tires have usable pressure and no obvious damage		
Brakes squeeze firmly and stop the bike during a roll check		
Handlebars and stem are tight		
Seatpost is secure		
Chain/drivetrain is present, routed, and not jammed		
Pedals/cranks are secure		

Electrical assist system

Item	P/F/R/N/A	Notes
Battery case has no swelling, heat, odor, or visible damage		
Battery is mounted securely		
Controller is mounted securely		
Motor cable is protected and not rubbing		
Connectors are seated and not damaged		
Wiring is clear of tires, chain, brakes, cranks, and pinch points		
Display/control unit is secure		
Throttle/PAS is not being touched during setup		

Charger is known, matched, and undamaged		

Rider and area

Item	P/F/R/N/A	Notes
Helmet available and fits		
Controlled area can be marked before any ride test		
No public traffic or uncontrolled pedestrian path		
Instructor has approved the next test step		

Part 5: Team risk report

Write one risk in each category.

Mechanical risk:

_____

Electrical risk:

_____

Rider-control risk:

_____

Area/environment risk:

_____

Exit ticket

1. What makes an e-bike different from a normal bicycle?

_____

2. What are three things that must be checked before any powered test?

_____

3. What is one thing you will check first tomorrow?

_____

Afternoon Lab

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Afternoon Lab: E-bike System Walk and Pre-Ride Inspection

Safe micro-mobility starts before the throttle moves.

Lab purpose

Students inspect an e-bike as a complete system and prove that safety comes before powered operation.

Safety boundary

- No riding today unless the instructor explicitly changes the plan after inspection.
- No throttle tests without instructor approval.
- No powered wheel spin without the bike stable, clear, and controlled.
- Battery remains off/disconnected unless the instructor directs otherwise.
- Any student can call stop-work.
- If something is unknown, damaged, loose, hot, swollen, rubbing, or unclear, mark it for instructor review.

Team roles

Role	Job
Inspector 1	Leads mechanical inspection
Inspector 2	Leads electrical inspection
Recorder	Writes pass/fail notes and risk report
Reporter	Shares findings during debrief

Rotate roles if time allows.

Lab A: Mechanical inspection

1. Stabilize the bike.

2. Identify the frame, fork, wheels, tires, brakes, drivetrain, cockpit, seat, pedals, and cranks.

3. Check for obvious looseness, damage, rubbing, missing parts, or unsafe setup.

4. Use the checklist in the student packet.

5. Mark anything uncertain as R for instructor review.

Lab B: E-bike system inspection

1. Identify battery, motor, controller, display, throttle/PAS, wiring, connectors, charger, and cutoffs if installed.

2. Look for damaged insulation, loose connectors, stressed wires, rubbing, pinch points, unsupported cable runs, and questionable battery condition.

3. Do not plug, unplug, charge, or power anything unless directed.

4. Use the checklist in the student packet.

Lab C: System flow map

Draw the basic flow:

Rider input -> controller -> battery power -> motor -> wheel motion

Then add safety checkpoints:

- brakes
- tire condition
- battery mount
- wiring clearance
- rider control
- area control

Lab D: Risk report

Each team reports:

1. One mechanical risk

2. One electrical risk

3. One rider-control risk

4. One thing that must be checked before tomorrow

Instructor scoring

Category	3 - Job-ready	2 - Acceptable	1 - Needs rework	0 - Unsafe/incomplete
Safety	Holds no-power boundary, uses stop-work thinking	Minor reminders	Frequent reminders	Unsafe action or ignores boundary
Inspection	Checks all major systems	Misses one minor item	Misses several items	Skips inspection
Accuracy	Correctly identifies bike/e-bike parts	Mostly correct	Confuses several systems	Cannot identify core systems
Documentation	Clear pass/fail notes and risks	Basic notes	Incomplete notes	No usable notes
Teamwork	Roles used well	Roles mostly used	Roles unclear	Disruptive or unsafe

Finish early extension

Have teams design a controlled ride-test area on paper. Include:

- start line
- stop line
- cone path
- observer zone
- emergency stop signal
- helmet/PPE check
- instructor checkpoint

Lab close

Before leaving:

- Tools returned
- Bike stable and powered off
- Packets collected
- Risks written on the board
- Tomorrow's first check identified

Google Classroom Post

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Google Classroom Post

Announcement

EBIKE101 starts today. We will treat e-bikes as real vehicle systems with mechanical parts, stored electrical energy, rider controls, and safety boundaries. Today's focus is not speed. Today's focus is proving the bike is safe before power or riding happens.

The nearly completed subwoofer from EDM101 may be shown in class as a project carryover, but it is not part of today's official EBIKE101 assignment.

Assignment title

EBIKE101 Day 1 - Introduction, Rider Safety, and System Inspection

Student directions

1. Complete the morning quiz independently.

2. Use the student packet during the bike and e-bike system walk.

3. Complete the pre-ride inspection checklist with your team.

4. Draw the basic e-bike system flow.

5. Complete the exit ticket before class ends.

Turn in

- Morning quiz
- Completed student packet
- Pre-ride checklist
- Exit ticket

Success criteria

- Identify the main bicycle and e-bike systems.
- Explain why inspection comes before powered motion.
- Complete a clear pass/fail safety checklist.
- Name one mechanical risk, one electrical risk, and one rider-control risk.

8-Day Curriculum Map

EBIKE101 Two-Week Curriculum Map

CTS Cohort 6

Dates: August 31, 2026 through September 10, 2026

Schedule: Monday through Thursday

Course spine

EBIKE101 teaches electric micro-mobility as a safe vehicle system. Students connect conventional bike mechanics, DC electrical safety, batteries, controllers, motors, rider input, conversion planning, wiring protection, troubleshooting, and final controlled user certification.

Official block coverage

- Rider safety and motor/throttle control
- Conventional bicycle mechanics and repair
- Electric bicycle conversion
- User riding certification

Two-week sequence

Day	Date	Focus	Main student artifact
Day 33	Mon Aug 31	Introduction, Rider Safety, and System Inspection	System map + pre-ride checklist
Day 34	Tue Sep 1	Bicycle Mechanics Foundations	Bike anatomy worksheet + repair log
Day 35	Wed Sep 2	Motors, Controllers, Throttle/PAS, and Signal Flow	Electrical control-path diagram
Day 36	Thu Sep 3	Battery, BMS, Charging, Connectors, and Electrical Safety	Battery safety checklist + connector inspection
Day 37	Mon Sep 7	Conversion Planning, Compatibility, Measurements, and JHA	Conversion plan + job hazard analysis
Day 38	Tue Sep 8	Mechanical Conversion and Component Mounting	Mounting inspection + photo documentation
Day 39	Wed Sep 9	Electrical Integration, Protection, and Troubleshooting	Wiring verification + fault log
Day 40	Thu Sep 10	Commissioning, Controlled Ride Test, and User Certification	Final commissioning sheet + user riding certification

Progression logic

1. Identify the system.

2. Inspect before operation.

3. Understand normal bicycle mechanics.

4. Trace electrical control and power.

5. Plan compatibility before conversion.

6. Mount components cleanly.

7. Wire, protect, and troubleshoot.

8. Commission and certify controlled use.

Safety doctrine

- No powered motion without a pre-ride inspection.
- No battery charging with unknown, damaged, or mismatched chargers.
- No wiring near moving parts, sharp edges, tire, brake rotor, chain, crank, heat, or pinch points.
- No public-road riding as part of class unless separately approved and controlled.
- Stop-work is professional behavior.
- Works, works safely, and works professionally are three different standards.

Labor Day note

Monday, September 7, 2026 is Labor Day. If AYA observes the holiday or the site closes, shift Days 37-40 to the next four class days. Do not compress conversion, wiring, and certification into unsafe time.